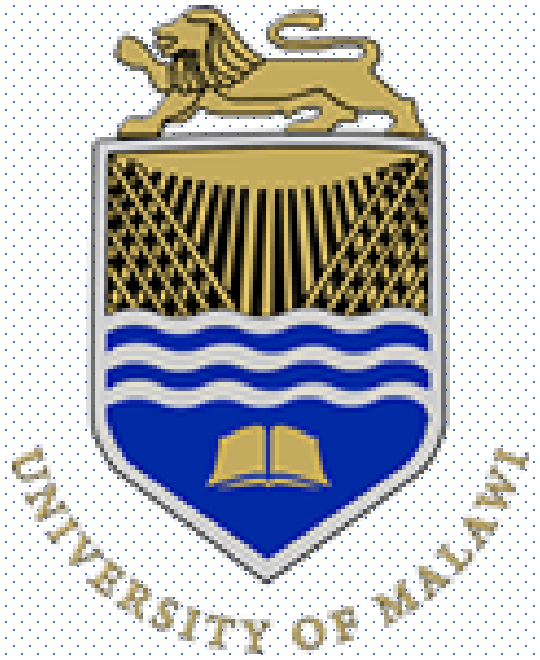
Web Design & Development

Modern Web Applications

Web Design & Development



Isaac S. Mwakabira

INF221 Web Design & Development?

- ✓The main aim is to:
  - ✓"Introduce you(**students**) to **modern** tools, methods, standards and technologies in ***website*** and ***web-application***: **design** and **development**"
  - ✓Largely practical, as such **the assumption & expectation** is for all of you to have done some basic computer programming – COM 121

Learning Outcomes

Students should be able to:

- ✓Analyze the structure of websites and web applications
- ✓Discuss and evaluate design tools, highlight compartibility issues sorrounding current web browsers
- ✓Build web applications, cross-browser compartible scripts and website adhering to set standards
- ✓Install and configure web servers
- ✓Deploy web applications

Indicative Content

✓The nature of the WEB – the World Wide Web(WWW)

The hypertext markup language(HTML), hyper transfer protocol(HTTP), and the basic HTML structure of the web page.

- Websites
- Webpages
- Webservers

✓Web Page Structure and Layers

✓Evolution of HTML

Indicative Content Cont…

- ✓ Cascading Style Sheets - CSS
- ✓ Web Development Processes
  - ➢ Understanding the business process
  - ➢ Defining the web project(s) goal(s)
  - ➢ Web development team
  - ➢ Web development phases
- ✓ Development Tools – collaboration, testing, debugging and others…
- ✓ Design Principles
- ✓ Frameworks

Indicative Content Cont…

- ✓Web server installations
  - ➢Development environments
  - ➢Production environments
- ✓Emerging Technologies – Containerization & Clustering(Docker, Kubernetes), Virtualization & many more
- ✓And many more…..

Assessment

- ✓ Continuous 50%
  - ➢ Individual Assignments - 20%
  - ➢ Group Assignment - 15%
  - ➢ Test 1 - 15%
  - ➢ Test 2 - ?
  - ➢ Total = (20 + 15 + 15 + ?) = 50%
- ✓ Final examination 50%

Class Logistics

- ✓?
- ✓Computer Labs X & LT1 and Lab Group A
- ✓Classroom code: shxgmrz

Learning - Methods and Activities

- ✓ Lectures – 2 hours per week
- ✓ Practical Lab Sessions – 2 hours per week
- ✓ Online Learning
  - ➢ Caution!!! No plagiarism, otherwise perpetrators will regret getting involved in this malpractice
- ✓ Projects – Both individual and Group

Topic schedule - *Incomplete*

- ✓Week 1 – Introduction & the nature of the Web (Standards, Protocols & etc.)
- ✓Weeks 2 to 3 – Web Page Structure and Layers; HTML, CSS & JS
- ✓Weeks 3 to 6 – Modern Javascript Language
- ✓Week 7 – Web Servers & Application Programming Interfaces(APIs)
- ✓Weeks 8 to 12 – Introduction to Web Frameworks
- ✓Week 13 – Web Design concepts and principles

Frameworks & Libraries

Languages

Databases & Servers

Others

But where do we start from?

An Ideal Web application architect or Software Engineer

Only through a fundamental understanding of the core technology can developers or engineers be expected to grow with the rapid technological changes associated with Web application development.

- Must be ***"the jack of all trades and master of all"*** *as opposed to* ***"the jack of all trades yet master of none"***

Web Design vs Web Development

- ✓Web Design:
  - ➢Web design encompasses many different skills and disciplines in the production and maintenance of websites or applications.
  - ➢The different areas of web design include **web graphic design**; **interface design(UI)**; **authoring**, *including standardized code and proprietary software*; **user experience design(UX)**; and **search engine optimization**.
  - ➢It is a discipline

Web Design vs Web Development

- ✓Web Development:
  - ➢**Web development** is the work involved in developing a website or web application for the Internet (World Wide Web) or an intranet (a private network)
  - ➢Actual work process involved in coming up with a website or web application i.e. writing code

The nature of the Web

- ✓ The essence of the Web: 3 basic components defines the web technology – Tim Berners-Lee
  - ➢ Markup language for formatting & cross-referencing hypertext document via hypertext links
  - ➢ Uniform resource notation for addressing accessible resources over the network (URI – Unified Resource Identifier or URL – Unified Resource Locator)
    - ❖ Generalized notation - URLs
    - ❖ **scheme://host**[**:port#**]**/path/.../**[**;url-params**][**?query-string**][**#anchor**] – *to be explored later on*
  - ➢ Protocol for transporting specialized messages over a network - core foundation of the WWW

The nature of the Web

- ✓ “The World Wide Web (known as "WWW', "Web" or "W3") is the universe of network-accessible information”
- ✓ The WWW is an information space where documents and other web resources are identified by Uniform Resource Locators (URLs, i.e. *https://www.example.com/*)
  - ➢ which may be interlinked by hypertext, and are accessible via the Internet”
- ✓ “The WWW is combination of all resources and users on the Internet that are accessible via the Hypertext Transfer Protocol (HTTP).”

The nature of the Web…

- ✓The World Wide Web is the universe of network-accessible resources, which may be interlinked by hypertext (connection of documents and information resources through links),
  - ➢And are accessible on the Internet using the ***Hypertext Transfer Protocol*** (HTTP), & other ***communication protocols***

Web pages to Web Sites

- ✓ The Web, as we know it today, can partially be attributed to its grass roots proliferation as a tool for personal publishing
- ✓ The technology was simple: a computer connected to the Internet, running a Web server, was all that was necessary to serve documents
- ✓ Web page vs. Web site
  - ➢ A Web site is more than just a group of Web pages that happen to be connected to each other through hypertext links
  - ➢ Web sites have *thematically-related* content which is *aesthetically* arranged to give a common look & feel to its pages. Maintaining content thematic consistency gives websites identity on the internet
  - ➢ Webpage – basic unit of information on the Web, containing texts, graphics, & other media

Web Applications

- ✓During its infancy, web sites served static content - found in *files*
- ✓Very few information services were truly considered *dynamic*
- ✓Dynamic information services:
  - ➢Services that connected the Web to relational databases enabled the web to become truly *dynamic*.
  - ➢No longer was it sufficient to say that you were designing a 'Website' (as opposed to a mostly collection of 'Web pages')
  - ➢It became necessary to designing & developing Web applications

Web Applications…

- ✓By definition, a web application is a **client/server** *application* that uses a Web **browser** as its **client program**, and performs an *interactive service* by connecting with *servers* over the *Internet* (or *Intranet*).
  - ➢A Web site simply delivers content from static files.
  - ➢A Web application presents **dynamically** tailored content based on **request parameters**, tracked **user behaviors**, and **security** considerations

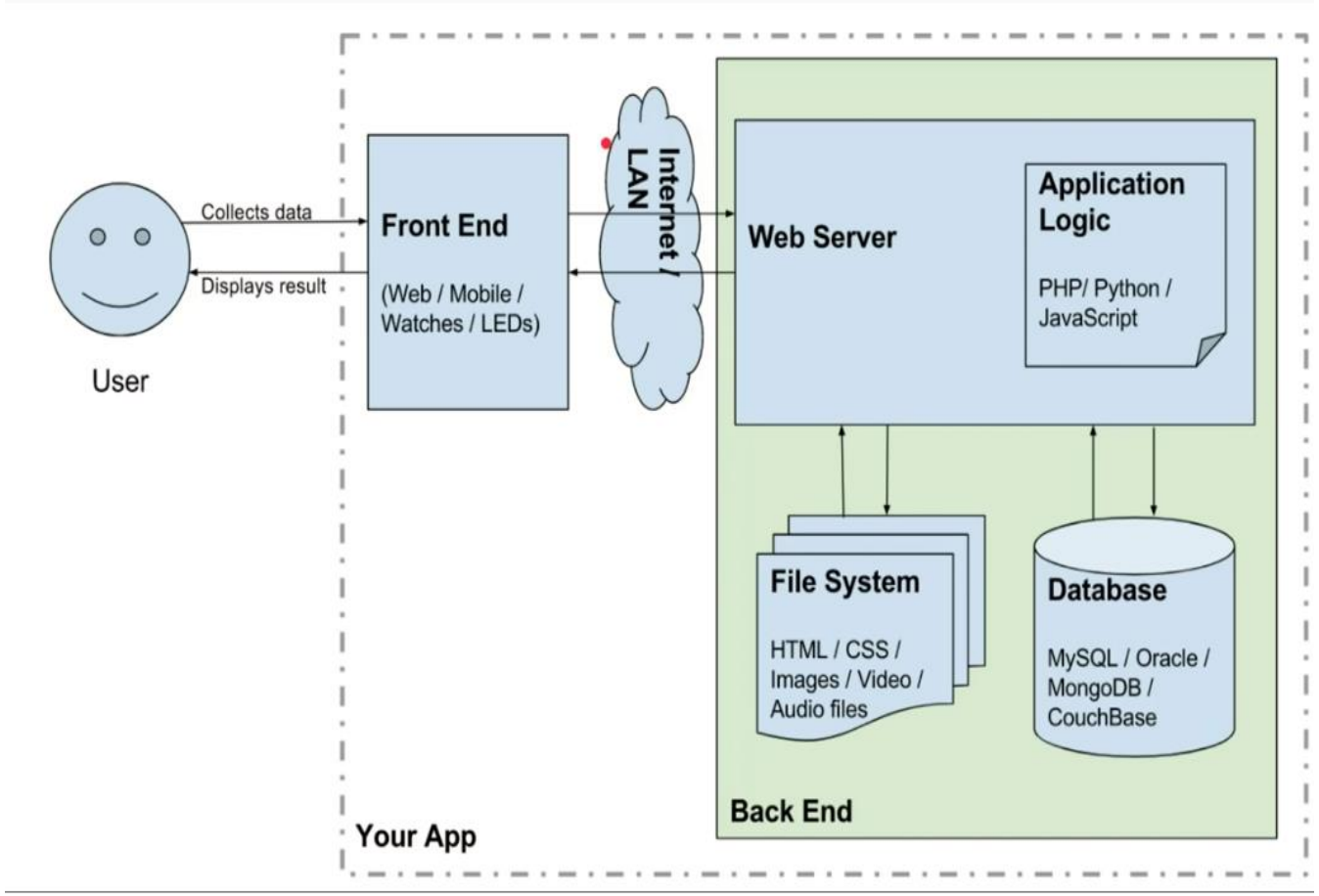
Website vs Web Application

- ✓Website:
  - A website is a collection of related web resources, such as web pages, multimedia content, which are typically identified with a common domain name, and published on at least one web server
  - Defined by their content and information
- ✓Web application
  - A **web application** or **web app** is a client–server computer program with the client running in the web browser. (Examples???)
  - Defined by interactions with the user and are typically action driven.

Web Protocols

Web Communication Protocols…

No communication is possible if “we” cannot speak the same language!!!!



Web Communication Protocols

- ✓ Set of rules that defines & governs the communication between multiple computers over a network
- ✓ TCP/IP – Transport Control Protocol/Internet Protocol
  - Transport layer – provides message transport services between remote applications running on remote computer systems,
  - Internet layer - provides the mechanisms for intersystem communications, controlling message routing, validity checking, and message header composition/decomposition. IP and Internet Control Message protocols (ICMP) operate on this layer
  - Application layer – mostly associated with “the Internet”
  - Network Interface layer – lowest level of data transmission within TCP/IP protocol stack

Web Communication Protocols…

- ✓SMTP – Simple Mail Transfer Protocol, UDP – User Datagram Protocol, IMAP – Internet Message Access Protocol, IM, POP – Post Office Protocol
- ✓IP Security(IPSec) – Used in Virtual Private Networks. Encrypts data sent between or among computers that share the same cryptographic keys
- ✓SSL & TLS
- ✓SOAP(Simple Object Access Protocol), XML-RPC(XML Remote Procedure Calls)
  - ➢ For Web services; requests return *machine-understandable* responses
  - ➢ Web services????

Web Communication Protocols…

- ✓ Advanced Message Queuing Protocol(AMQP)
  - ➢ Message orientation, queuing, routing(point-to-point or pub/sub), reliability, security
- ✓ Streaming Text Oriented Messaging Protocol(STOMP)
- ✓ Message Queuing Telemetry Transport(MQTT) –
  - ➢ Lightweight pub/sub protocol. Runs on TCP/IP
  - ➢ Includes a **broker** server and several **clients**
  - ➢ RabbitMQ is an implementation of this protocol plus others(STOMP, AMQP)

Exercise

- ✓ List & discuss any Message Queuing Protocols & Services(like JMS) existing today, while listing advantages and disadvantages for each protocol.

Web Communication Protocols…

- ✓ FTP
  - File Transfer Protocols – offered the universal medium of transferring and sharing resources or information using computer networks on the internet
  - Command line based- not very user friendly by today's standards
- ✓ *FTP* + *hypertext* spearheaded the current advancements in web technologies – HyperText Transfer Protocol
- ✓ HTTP (and its secure variants S-HTTP and HTTP-S)
  - Is the standard protocol for the communication between a browser and a Web server
  - Specifies the group of guidelines that allow the exchange of information on the Web
  - HTTP is a stateless protocol, its usage is restricted in some way
  - **Stateful** (session-based) vs. **Stateless** protocols? Discuss? Examples???

Web Communication Protocols…

✓Web Sockets

- Two-way communication protocol, provides a full duplex connection
- Websocket vs. HTTP – when does each close the connection?
- Communication is initially established over HTTP(Handshake), any subsequent communication is upgraded to TCP protocol.
- Both server or client can send messages to each other, after the upgrade – applicable in real-time applications
- Any of the two machines involved in the communication is allowed to close the connection

Web Communication Protocols…

- ✓The web is built on top of core internet communication protocols
  - ➢Understanding the **relationship** between the ***"Web technology"*** and the underlying protocols is key to the design, development & implementation of true ***"Web applications"***
  - ➢Exploiting such relationship help us make clear distinctions between *"Web pages"* or *"Websites"* and *"Web applications"*

Architectures

Software Architecture

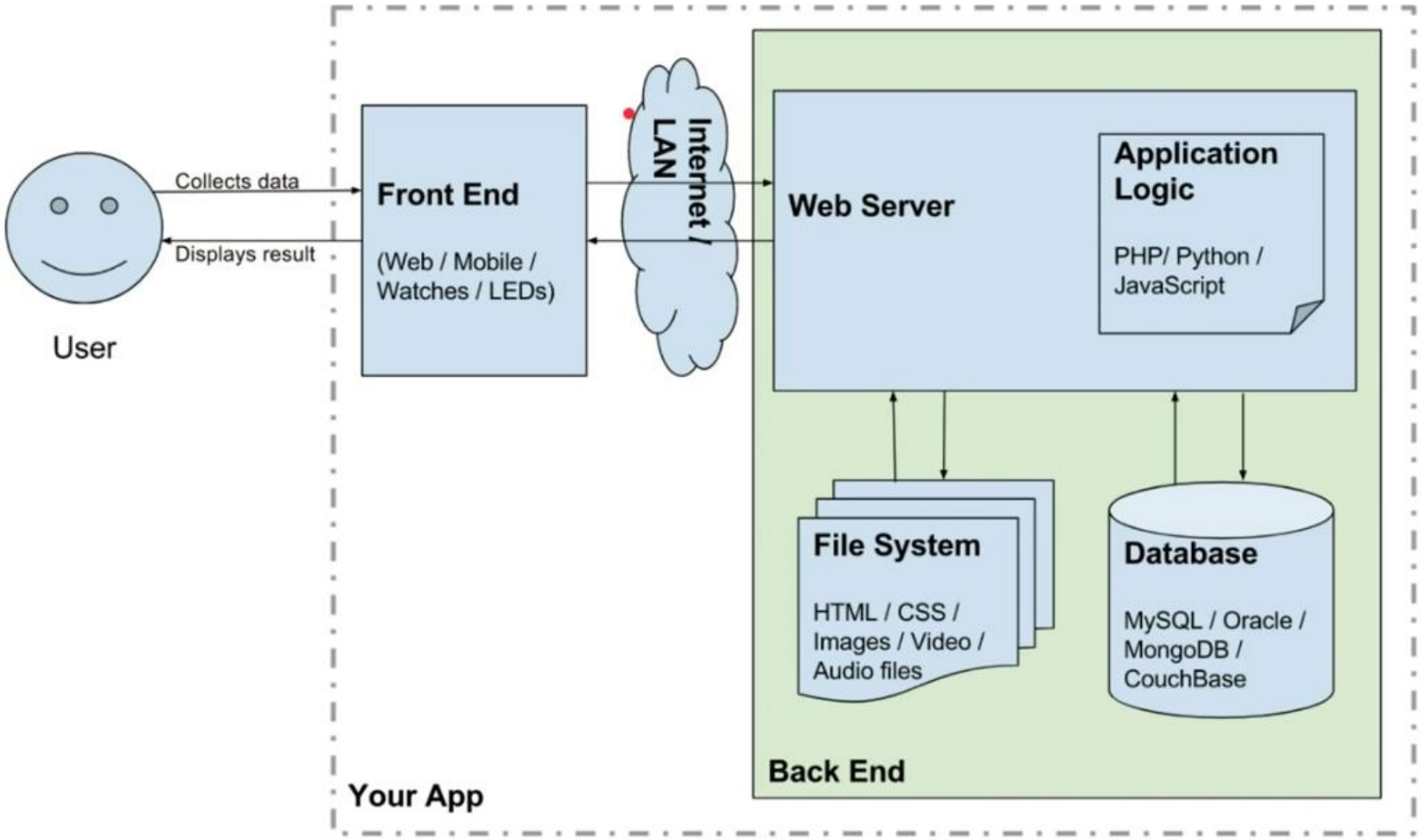
- ✓ Definition:
  - ➢ Means defining an application in terms of its computational components & specifying how these components are related & behavioral interactions among themselves to achieve overall system goals
  - ➢ These components/elements may be grouped into larger subsystems
  - ➢ Main goals: resilience, usability, performance, reuse, functionality, comprehensibility etc.

Software Architecture…

- ✓ Server-client
- ✓ Multi-tier
- ✓ Network and distributed – N-tier
  - ➢ Service-oriented(SOA) – Web services
    - ❖ Principles: Interoperability, location transparency, discoverability, loose-coupling & encapsulation, abstraction, autonomy, statelessness, standard interfaces & contracts, reusability, dynamic configuration
  - ✓ Cloud architectures
    - ❖ Software as a Service (SaaS)
    - ❖ Platform as a Service (PaaS)
    - ❖ Infrastructure as a Service (IaaS)

Software Architecture…

- ✓Standards & protocols employed to developing web services
  - ➢XML, SOAP, REST (**RE**presentational **S**tate **T**ransfer), JSON
- ✓Server-client architectures
  - ➢Server
  - ➢One to many clients


User
Collects data
Displays result
Front End
(Web / Mobile / Watches / LEDs)
Internet / LAN
Web Server
Application Logic
PHP/ Python / JavaScript
File System
HTML / CSS / Images / Video / Audio files
Database
MySQL / Oracle / MongoDB / CouchBase
Back End
Your App

How does the web work????

CLIENT

Program used to view web page

How does the web work????

CLIENT

Program used to view web page

SERVER (cc.ac.mw)

Stores and serves a website content

How does the web work????

CLIENT

Program used to view web page

Request: GET
cc.ac.mw

SERVER (cc.ac.mw)

Stores and serves a website content

How does the web work????

Request: GET
cc.ac.mw

Response: cc.ac.mw webpage files

CLIENT

Program used to view a webpage

SERVER (cc.ac.mw)

Stores and serves a website content

Assignment 1

- ✓Define “version control management system”.
- ✓Discuss why version control management systems are important to software development. For each type of version control system you discuss, give two examples and discuss any underlying fundamental concept(s) behind them.

Client Side &
Server Side Scripting

Differences

- ✓ Client side:
  - The user's device browser is used to run scripts
  - The processing takes place on the end users computer
  - The source code is transferred from the web server to the users computer over the internet

Differences…

- ✓Server side - runs a scripting language on a **web server**;
  - ➢A user's request is fulfilled by running a script directly on the **web server** to generate **dynamic** HTML pages. This HTML is then sent to the client **browser**
  - ➢It is usually used to provide interactive web sites that interface to ***databases*** or other ***data stores*** on the server

Differences…

- ✓Server side
  - ➢The primary advantage to server-side scripting is the ***ability to highly customize the response*** based on the ***user's requirements***, ***access rights***, or ***queries into data stores***

Web standards

Web Standards…

- ✓ Definition:
  - ➢ Core set of ***rules*** for developing ***websites or web applications*** that may be accessed ***across*** different ***device configurations & environments***
  - ➢ Technologies used to **create and interpret** web-based content – established by **standards** bodies like the W3C(others???). They are designed:
    - ❖ to future-proof documents published on the Web
    - ❖ to enable the accessibility of documents to as many people and devices as possible.
    - ❖ to enable developers or engineers design applications that are **functional**, **accessible** and **compatible across** various computer environments

Web Standards…

- ✓Importance;
  - Accessibility & Adaptation
  - SEO – Search Engine Optimizations
  - Backward compatibility
  - Separation of concerns

Web Standards…

- ✓World Wide Web Consortium (W3C) is the main central body behind the setting of these standards
- ✓Structural languages – HTML, XHTML, XML
- ✓Presentation languages – CSS(Level 1, 2, 3 etc.), SVG (Scalable Vector Graphics)
- ✓Behavior – defined by the ECMAScript(ES), Describes a general purpose or standardized version of a programming language - Javascript language
- ✓Document Object Model – DOM(Core & HTML)

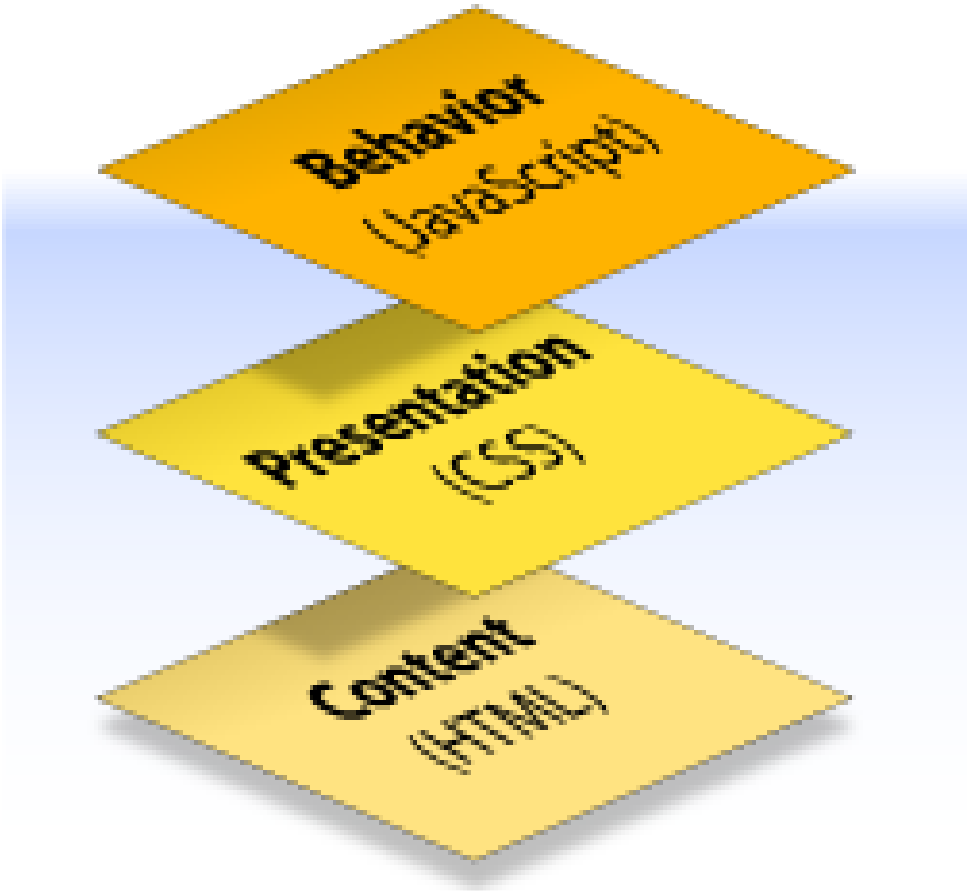
Web Standards…

- ✓ Document Object Model:
  - ➢ It is an application programming interface (API) for HTML & XML documents
  - ➢ Defines the logical structure of documents – specifies how they may be accessed and manipulated
  - ➢ The DOM is designed to be used with any web programming language – language independent
  - ➢ Abstract interfaces are specified for accessing and manipulating the internal representations of a document
- ✓ The main objective is to provide a standard programming interface – for usage in various environments & applications – W3C
- ✓ How do we know a web page or document adheres to set web standards?

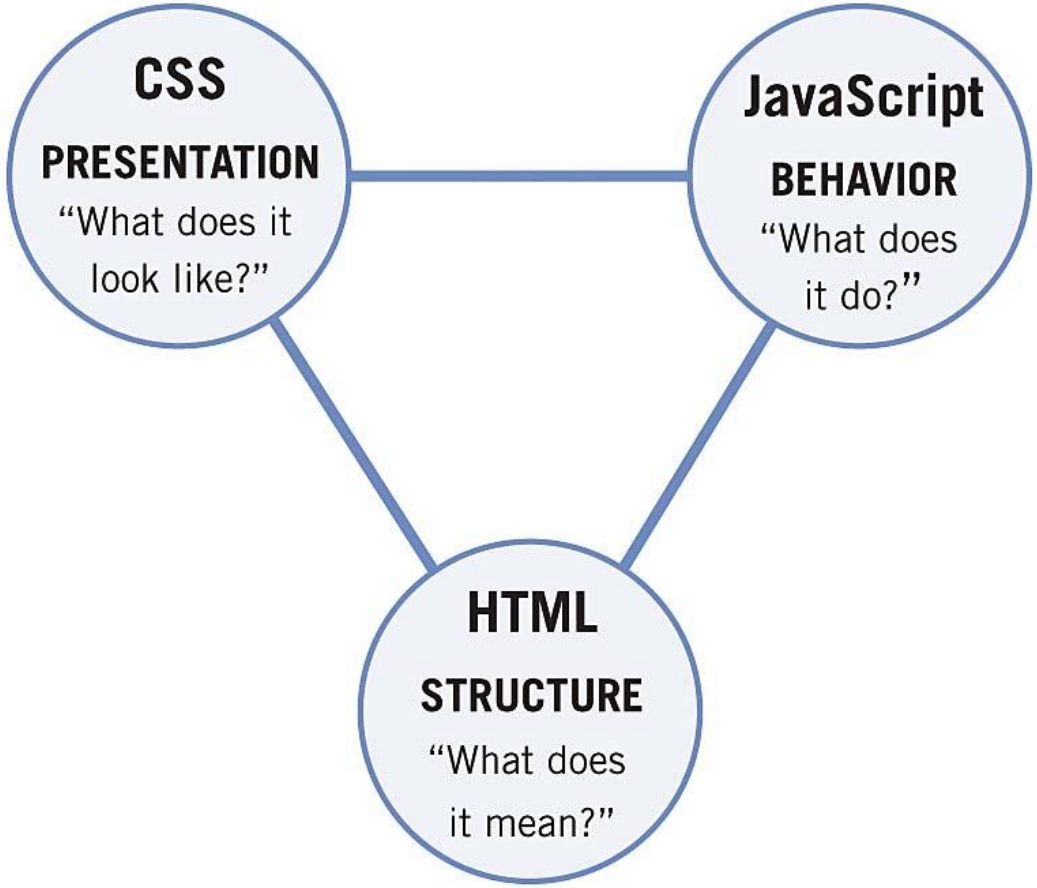
Exercise

- ✓ Find any three websites or applications(Open them in your favorite browser, right-click on each site and click "**view page source**" on the pop-up menu), then note the following:
  - Whether client-side or server-side scripting was used during development
  - Whether they are websites or web applications
  - Whether any of the chosen sites followed the set web standards discussed during development
- ✓ Justify your reasoning

Web Page Layers and Structures



Web Page Layers



Web Page Layers

- ✓Content - (X)HTML - Structure
  - Defines the content &/or structure of a document as specified by the DOM
  - Mandatory parts of the document – HTML. Semantic & Structured markup of a document
  - Each & every web browser validates this structure before it is rendered
  - Extensible HTML has its rules more stricter than HTML. It is a rewrite of HTML as an XML language
    - I.e. case sensitivity, closing of tags

Web Page Layers…

- ✓ Content - (X)HTML - Structure
  - ➢ XML
    - ❖ “meta-language” – used to create & define other mark-up languages
    - ❖ Examples GML, RSS
  - ➢ Since many of these markup languages were written in XML, their contents can be shared across various applications that are using variants/implementations of XML.

Web Page Layers…

- ✓ Presentation (CSS)
  - Look and feel
  - Dictates how the structured HTML document should look like
  - It also include the use of **media queries** that changes the way web applications render across devices of different sizes
  - SASS, LESS etc. - pre-compilers

Web Page Layers…

- ✓ Behavior(JS)
  - Layer of interactivity and dynamic *behavior* – manipulates DOM elements via exposed interfaces
  - Programming languages – JS, Coffee script, Java, Swift, PHP, Python, Ruby, Objective – C, etc.

Web Page File - Document

HTML

```
<body>
    <div>
        <p>So this is INF 221? Noice!!</p>
    </div>
</body>
```

CSS

```
<style>
    body {
        background: #dbdbdb;
        color: #ffffff;
        font-family: "Comic Sans"
    }
</style>
```

JS

```
<script>
    var className = 'INF 221'
    console.log(className)
</script>
```

Introduction to HTML & CSS

Web Page Layers…

- ✓HTML
  - Contains text, tags gives structure to the text
  - Tags describe such things as ***links***, ***paragraphs***, ***headings, inputs, forms*** etc
  - Simply put, it is a set of nested boxes.
  - These boxes are referred to as ***elements.***
- ✓HTML Element
  - An HTML element is an individual component of an HTML document/web page.
  - Used to structure content of an HTML document. Represented by HTML ***tags***

Exercise – *Developmet Tools*

- ✓Download and install Visual Studio Code (VS Code)

- ✓This **MUST** be done before coming to Friday's lab session

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In December, 2018, students engaged in the 16for16 campaign, which was part of the global 16 Days of Activism. The campaign was led by Limodzi, which is a groupin...

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Web Page Layers – HTML Tags…

✓Tag

- Defines HTML elements, thus giving structure to the document
- An element usually consists of a ***start*** tag and an ***end*** tag, with the content inserted in between.
- Example: My site paragraph text goes here…
- The HTML element is everything from the start tag to the end tag, including its content

Web Page Layers – HTML Tags…

<html>	Root of HTML Document
<head>	Info about Document
<body>	Document Body
<h1>, <h2>, <h3>, ...	Header tags
<p>	Paragraph tags
<ul>, <ol>	List tags
<img>	Image tags
<a>	Anchor (link) tags
<form>	Supports input
<header><aside>...	Container (group) tags, defines section
<script>	Loads JavaScript
<style>	Loads CSS styles

Web Page Layers – Tags…

- ✓ Attributes
  - Opening ***tags*** can specify ***attributes***, which tell us more about the content of that element.
  - ***Attributes*** require a ***name*** and a ***value***.
- ✓ ***Note***
  - To learn HTML you need to know what tags are available for you to use, what they do, and where they can be used

Web Page Layers – Tags…

```
<p lang="en-us">Paragraph in English</p>
```

Attribute name

Attribute value

Web Page Layers – The DOM…

- ✓Structure
  - ➢The Document Object Model (DOM) is a programming API for HTML (and XML) documents
  - ➢Hierarchy of HTML elements
  - ➢The Core interface defines the elements access methods that are implemented by HTML and XML documents

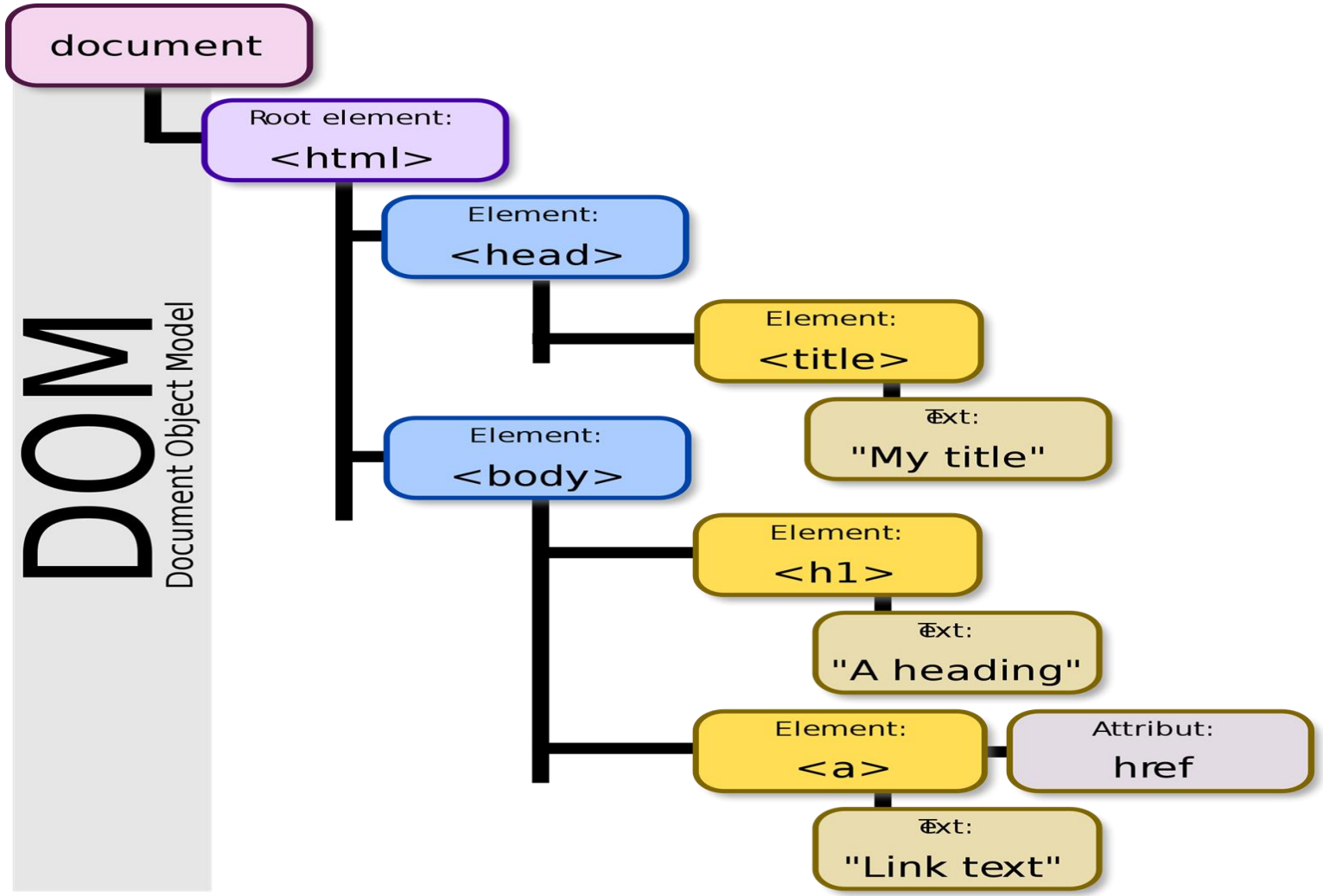
Web Page Layers – The DOM…

- ✓Structure
  - ➢It defines the logical structure of documents and the way a document is accessed and manipulated
  - ➢With the Document Object Model, programmers can create and build documents, navigate their structure, and add, modify, or delete elements and content.

Basic Structure – The DOM…

```html
<!DOCTYPE html>
<html>
    <head>
        <title>My Title</title>
    </head>
    <body>
        <h1>A heading</h1>
        <a href="http://www.example.com">Link Text</a>
    </body>
</html>
```

Web Page Layers – The DOM…



Web Page Layers – The DOM…

- ✓All HTML documents ***must*** start with a document type declaration
  - <!DOCTYPE html>.
- ✓HTML ***documents*** begins with <html> and ends with </html>
- ✓The ***visible*** part of a web page is rendered between <body> and </body>

HTML CONTENT MODELS

- ✓ Define the type of content expected inside an element and control syntax rules such as element nesting etc.
  - ❖ Metadata content
  - ❖ Flow content
  - ❖ Sectioning content
  - ❖ Heading content
  - ❖ Phrasing content
  - ❖ Embedding content
  - ❖ Interactive content

```html
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <title>Page Title</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">
</head>
<body>
    <h1>My home site</h1>
    <p>
        My site paragraph <a href="http://www.example.com">Link</a>
    </p>
</body>
</html>
```

My home site

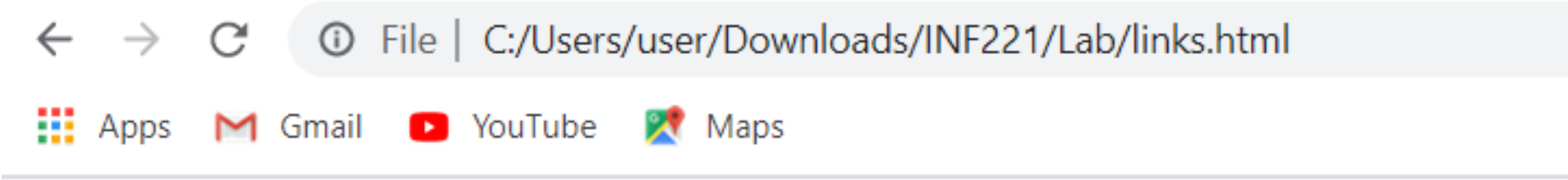
My site paragraphLink

Examples – Inserting Links…

Lab ▸ links.html ▸ html ▸ head

```html
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <title>Inserting Links</title>
</head>
<body>
    <h1>Link Example</h1>
    <p>
        Hello, here is a
        <a href="https://bit.ly/2U9dF8L" title="Github course material">Link</a>
        to course material.
    </p>
    <p>
        And here is a
        <a href="./index.html" title="Index html document">Link</a>
        to index.html document.
    </p>
</body>
</html>
```

Examples – Inserting Links…



Link Example

Hello, here is a Link to course material.

And here is a Link to index.html document.

More Examples– Tags…

- ✓Lists – ordered and unordered
- ✓Inserting images
- ✓Form elements

More Examples– Forms…

✓Form elements

- <input type="text"> Text input
- <input type="radio"> Radio input
- <input type="submit"> Submit button
- <input type="search"> Search input
- <fieldset><legend> Group form elements

✓ <form action="" name="" class="" id="">your form goes here</form>

CSS – Styling HTML Document

- ✓Cascading Style Sheet
  - Language that describes how HTML elements are to be displayed on screen, paper, or in other media
  - List of rules that tell your web browser how content should look
- ✓CSS rules affect how HTML boxes look

Without CSS

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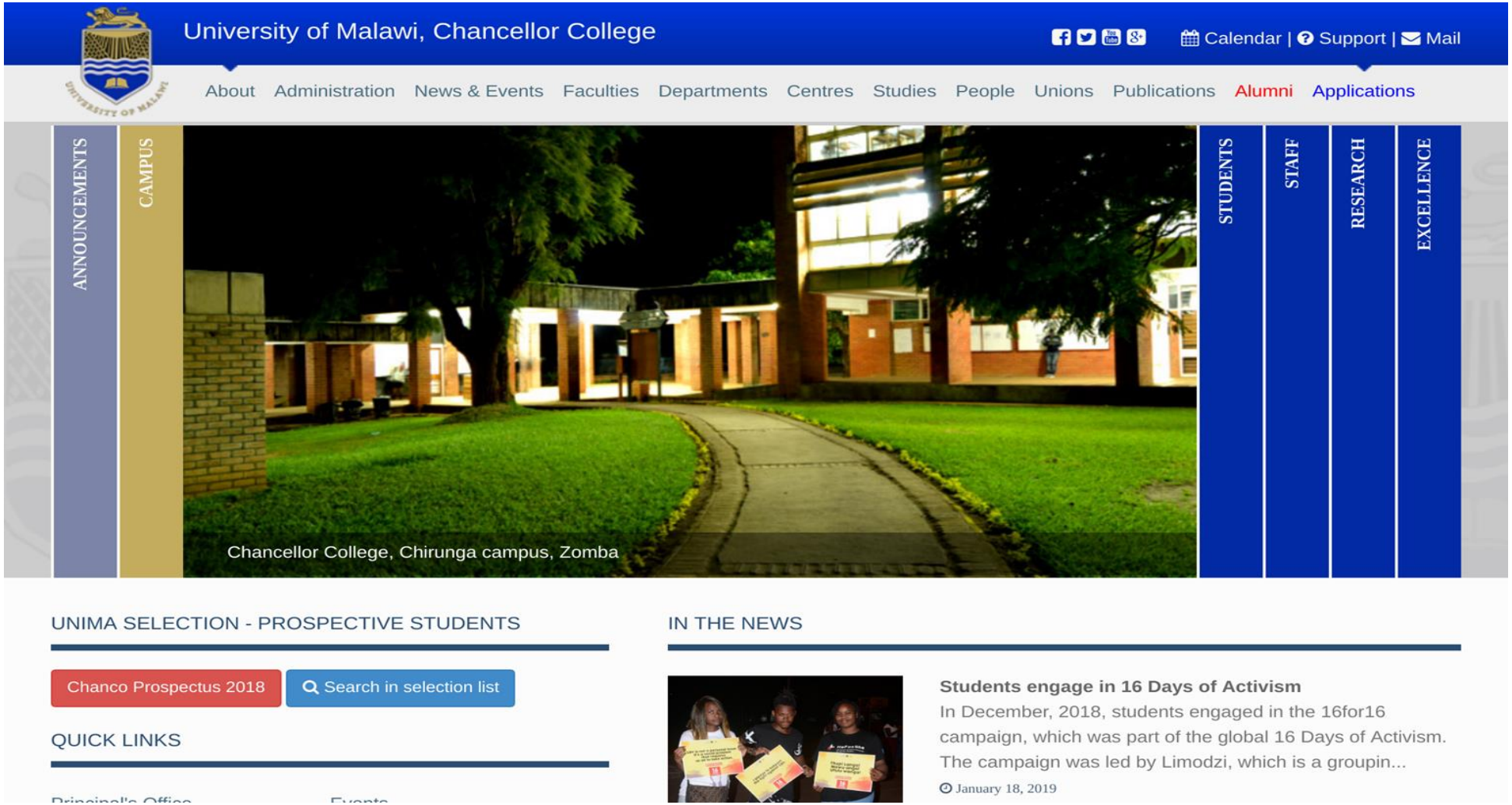
  - Bachelors Degree Studies
  - Masters Degree Studies
  - Doctorate Degree Studies
  - Other Studies

With CSS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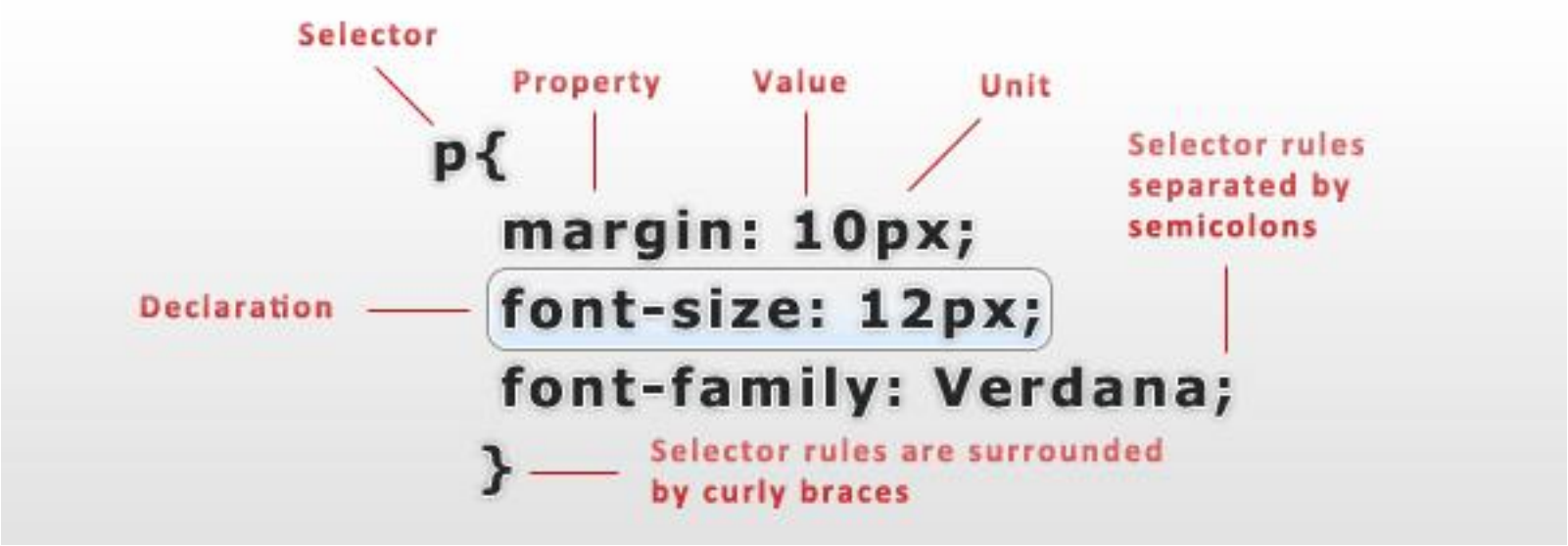
IN THE NEWS
Students engage in 16 Days of Activism
In December, 2018, students engaged in the 16for16 campaign, which was part of the global 16 Days of Activism. The campaign was led by Limodzi, which is a groupin...
January 18, 2019

CSS Anatomy

```
selector      {
  property: value;
}
```



Data Exchange Formats

Data Exchange Formats

- ✓ CSV - so far the most used data import & export for spreadsheets and some databases - excel
- ✓ JSON(Javascript Object Notation) - scalars: (“strings”, true or false, null, numbers) & structural characters: ({}, [], **:** & **,**)
- ✓ GeoJSON
- ✓ XML – stores hierarchical data. Can be represented using a graph tree.
  - ➢ Primary use: **RPC** – making remote procedural calls, **object serialization** for data transfer between applications, defining & specifying other markup languages etc.
- ✓ YAML – superset of JSON, designed to be human readable and commonly used for configuration files
- ✓ GML – Geography Markup Language

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Resources

- ✓Ducket, J (2011). *HTML and CSS: Design and Build Websites*. Indianapolis, USA
- ✓W3 Schools - https://www.w3schools.com/
- ✓Mozilla developer network - https://developer.mozilla.org
- ✓WHATGP - https://html.spec.whatwg.org
- ✓Web Platform - https://webplatform.github.io

Resources Cont…

- ✓Journal – international Journal of Web Engineering and Technology(IJWET)
- ✓https://developers.google.com/web/fundamentals/?hl=en